Referee report on the rank-two toral positivity note

(with Part IV, Draft 5, and its revision log)

Claude (Anthropic), for David Joyner

November 7, 2026

0. What was submitted

Three files were submitted.

- `mjl_part4_draft5.tex`. This is byte-for-byte identical to the Draft 5 I refereed earlier today (same MD5 checksum). My earlier report on it stands. In particular, delete the nongood-class hypothesis in Proposition 8.5, correct the description of B_d, and label Lemma 6.4.
- The Markdown revision log for Draft 5. It is accurate. The Python check script that it mentions was not attached.
- The new note, *A permutation-character theorem for quadratic wild-place discriminant ratios.* This is the substance of this report.

I read all three files in full.

1. The note: verdict

The theorem is correct. I checked every step by hand (§1.1). I also wrote an independent script that recomputes the coefficients c_k from the valuation data. It covers $p \in \{3, 5, 7\}$, $p^a \in \{p, p^2\}$, several ramification indices and residue degrees for F, and both types of E: 72 cases in all. In every case every c_k is a nonnegative integer, and the telescoped sum equals $R_E(1) = |n|_F^{-1}$. The examples are also right: $1 + 104\,\mathrm{Reg}(C_6)$ and $1 + 12\,\mathrm{Reg}(C_2) + 60\,\mathrm{Reg}(C_{10})$ for $\mathbb{Q}_5(\zeta_5)$ both have value $625 = |5|_F^{-1}$ at the identity.

This settles Step T of Part IV, §6.4, for $r = 2$, at every wild place and for every odd n. Together with Lemma 6.4 it implies the following. For every ordinary irreducible ρ of a quaternion division algebra, the restriction of $L_D\Theta_\rho$ to every covered maximal torus is an actual genuine character.

1.1 Checks

- *The discriminant.* $\Delta(t) = |N(1-u)|_F^{1/2} = |1-u|_E^{1/2}$, since $\mathrm{Ad}(t)$ acts on $D/E = Ej$ by multiplication by $u = t/\sigma(t)$. Then $1 - u^n = \prod_{\zeta \in \mu_n}(1 - \zeta u)$.
- *Roots of unity with nontrivial prime-to-p part.* For these, $|1 - \zeta u|_E = 1$. The reduction of $u \in E^1$ meets $k_F^\times$ only in $\{\pm 1\}$: in the unramified case because the norm-one group of $k_E^\times$ meets $k_F^\times$ in $\{\pm 1\}$, and in the ramified case because $\bar{u}^2 = 1$. A nontrivial root of unity of odd order is not ± 1.

- *The radial formula.* $v_E(1-\zeta u) = \min(v_E(1-u), t_\zeta)$. The equality case is excluded by applying σ, using $\sigma(u) = u^{-1}$, and $v_E(\zeta - \zeta^{-1}) = t_{\zeta^2} = t_\zeta$, which holds for p odd.

- *Value at the identity.* $\prod_{\zeta \in \mu_{p^a} \smallsetminus 1}(1-\zeta) = p^a$ gives $R_E(1) = |n|_F^{-1}$. This agrees with Part III, Lemma 3.3, for $r = 2$.

- *Filtration indices.* Unramified: $[G : G_k] = (q+1)q^{k-1}$. Ramified: $[G : G_1] = 2$ and $[G : G_{2j}] = 2q^j$, with all positive depths odd. The ramified case uses the parametrization $c \mapsto u$, and $u - u'$ has E-valuation $2v_F(c - c') + 1$.

- *Integrality of c_k.* It uses three facts:

  - the exponents s_k are even, so $q+1$ divides $q^{s_k} - 1$;
  - q^{k-1} divides M_{k-1}, respectively q^j divides M_{2j-1};
  - M_1 is odd.

  All correct.

1.2 Minor repairs to the note

(1) *Surjectivity at each level.* The unramified filtration indices are stated without proof that the norm-one group surjects onto the trace-zero residue groups at every level. Add one line, via Hensel's lemma, or via the Hilbert 90 parametrization as in the ramified case.

(2) *Where p odd is used.* The theorem uses p odd in two places: $\zeta \neq \zeta^{-1}$, and the $\{\pm 1\}$ argument. Say so in the statement.

(3) *Labels.* Give the theorem a **[P]** label, as in the series.

(4) *Integration.* Make it Proposition 6.6 of Part IV, replacing the example-level Proposition 6.5. Then state the consequence for *all* ρ via Lemma 6.4.

(5) *Its last remark.* It is correct: positivity of one row does not give integrality of that row. This exposes an overstatement in my Draft 5 report (§2).

2. A correction to my previous report

In my Draft 5 report, §2(ii), I said the positive-semidefinite test for the single row $\rho = \mathbf{1}$ would settle that case. That is too strong.

- Positive-semidefiniteness of one row shows only that the row is nonnegative.

- The positivity lemma needs *all* rows to be nonnegative, because disjointness of supports comes from comparing rows.

The finite test is still useful: a negative eigenvalue would disprove integrality at once, and it is cheap. But to *prove* irreducibility of $L_D\mathbf{1}$ one needs either integrality of that row, or nonnegativity of every row whose support meets that of $L_D\mathbf{1}$. The lemma itself, (a)⇔(c) for the whole matrix, is unaffected.

3. Can the wild integrality problem now be solved in general?

Not yet. But it can now be stated much more precisely. Here is where things stand.

What is proved.

- The unitary transform and its matrix (Part III).
- Integrality $\Leftrightarrow$ nonnegativity of the whole matrix (the positivity lemma).
- Torus-restriction positivity for every ρ when $r = 2$ (this note together with Lemma 6.4).

What is not proved.

(a) *Step T for $r \geq 3$.* The note's argument depends on $\sigma(u) = u^{-1}$ for norm-one elements of a quadratic extension. For degree $r \geq 3$ the root ratios $u_{ij} = \lambda_i/\lambda_j$ lie in the Galois closure, and non-Galois E occur. The equality case of the radial formula and the filtration indices both need new arguments.

(b) *Step G, gluing, for every r including $r = 2$.* This is the real obstruction. Positivity on every torus does not imply positivity on the full group. Positive-definiteness is not determined by restrictions to abelian subgroups.

3.1 What torus positivity does give: rationality, via Artin

The following is a short consequence that sharpens what Step G must supply. I believe it is correct, but it has not been checked by anyone else.

Proposed Corollary (**[P]**, given Step T on all maximal tori; in particular for $r = 2$)**.** *Every coefficient $a_{\rho\tau}$ is rational, with denominator dividing $|Q|$ for any finite twisted quotient $Q = \widetilde{H}/N$ through which $L_D\Theta_\rho$ and Θ_τ factor.*

Proof sketch. (i) *Cyclic subgroups lie in covered tori.* Every cyclic subgroup of Q is the image of some $\langle g\rangle$ with $g \in \widetilde{H}$. Since r is prime, $p(g)$ lies in a maximal subfield E, so $\langle g\rangle \subset \widetilde{E^\times} = p^{-1}(E^\times)$.

(ii) *The restriction to a torus is a character of its image.* By Step T, $\varphi = L_D\Theta_\rho$ restricts to an actual character of $\widetilde{E^\times}$. This character is invariant under translation by $K = \widetilde{E^\times} \cap N$. For a nonnegative combination of characters, $\varphi(k) = \varphi(1)$ forces k to act trivially in every constituent. So φ restricts to a genuine character of the image of $\widetilde{E^\times}$ in Q, hence of every cyclic subgroup.

(iii) *Artin's induction theorem.* It gives $|Q| \cdot 1_Q = \sum_C a_C \operatorname{Ind}_C^Q \theta_C$, with $a_C \in \mathbb{Z}$ and θ_C ordinary virtual characters. Multiplying by the genuine function φ and using $\varphi \cdot \operatorname{Ind}_C \theta = \operatorname{Ind}_C(\varphi|_C\, \theta)$ gives $|Q|\varphi \in R^{\mathrm{gen}}(Q)$.

□

The local version. The same multiplication trick turns the local form of Brauer's induction theorem into a criterion. For each prime ℓ there is an integer m prime to ℓ with $m \cdot 1_Q$ a $\mathbb{Z}$-combination of inductions from ℓ-elementary subgroups (cyclic of order prime to ℓ, times an ℓ-group). So

$$a_{\rho\tau} \in \mathbb{Z}_{(\ell)} \iff \varphi|_E \text{ is a virtual genuine character for every } \ell\text{-elementary } E \leq Q.$$

This twisted version needs no new theory. It follows formally from the ordinary induction theorems, which answers the hedge in my previous report.

What remains. Integrality is now a prime-by-prime problem.

- $\ell \neq p$. The ℓ-groups map injectively to the finite metacyclic quotient H/ZU_D^1, so the ℓ-elementary subgroups are small. They are plausibly controlled by tori and the tame (Adams-type) arguments. But there is a gap: an abelian subgroup of Q is only commutative modulo N, so it need not lie in one torus. This needs an argument.
- $\ell = p$: *the core.* The p-elementary subgroups $C \times P$ contain nonabelian p-subgroups P of the covered pro-p group $\widetilde{U_D^1}$. Over that group the cover does not split when $p \mid n$ (Part III, Proposition 8.1). This is exactly where no torus-level information can suffice.

3.2 The most promising routes to the p-local core

(1) *Explicit construction for $r = 2$.* D is quaternion and p is odd, so D is tamely ramified. Its representations are explicit (Howe's admissible pairs; Corwin–Howe characters).

- Build the genuine representations $\tau(E, \tilde{\xi})$ by inducing from $\widetilde{E^\times} \cdot \kappa^*(U_D^m)$, using the Stone–von Neumann module on the torus.
- Compute their characters by the Frobenius formula.
- Match them with $L_D\Theta_\rho$. The note's explicit indicator form of R_E is exactly what such a comparison produces.
- By the positivity lemma it is enough to show that each $L_D\Theta_\rho$ is the character of *some* representation.

In my judgment this is now the most realistic path to a complete theorem for $r = 2$.

(2) *Kirillov theory on the deep pro-p part.* On deep congruence subgroups, where the cover splits, the power map is dilation by n, and dilation maps coadjoint orbits to orbits. The dimension count gives exactly c_D (my previous report, §3.4). This targets precisely the p-elementary subgroups above.

(3) *The global route.* This is the route of Part IV, Proposition 8.5, combined with the positivity lemma. It remains conditional on the split-place and complex-place matching.

(4) *Computation.* Compute the $\ell = p$ restriction test, or the full row, for $(2, 3, \mathbb{Q}_3(\zeta_3))$ at the first few levels. A failure would be decisive. Success would indicate what the general construction must look like.

Bottom line.

- $r = 2$. Step T is finished, and by the corollary of §3.1 the coefficients are rational. The remaining gap is the p-local part of gluing, which an explicit construction should be able to close.
- $r \geq 3$. Step T itself is still open.
- *In general.* I would not describe the problem as solvable "now". It has been reduced to one clearly identified local question for $r = 2$, and to two questions for $r \geq 3$.

4. Suggested next revision of Part IV

(1) Make the changes from my previous report (Proposition 8.5, B_d, labels).

(2) Insert the note's theorem as Proposition 6.6 **[P]**, with the consequence for all ρ.

(3) Add the positivity lemma, with the single-row caveat of §2.

(4) Add the Artin rationality corollary and the ℓ-local criterion as **[P]**, after an independent check.

(5) Restate the open problem as: *p-local integrality on p-elementary subgroups of the covered pro-p group* (for $r = 2$), plus Step T (for $r \geq 3$).